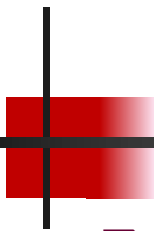


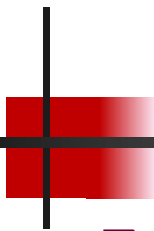
ElGamal Cryptosystem



ElGamal Cryptosystem

- ❑ The **ElGamal Cryptosystem** is an asymmetric key encryption algorithm
 - based on the **Discrete Logarithm Problem**
- ❑ It is widely used in **cryptography** for secure communication.

Components of ElGamal Cryptosystem



- ElGamal consists of three main stages:
 - Key Generation
 - Encryption
 - Decryption

Key Generation (Public and Private Keys)

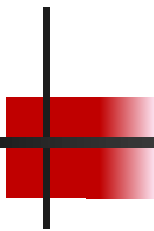
- ❑ A user generates a **public-private key pair** as follows
- ❑ Choose a large prime number p (a prime number used for modular arithmetic).
- ❑ Select a primitive root e_1 modulo p (i.e., a generator of the cyclic group).
- ❑ Pick a random private key d where d is a secret integer
$$(1 < d < p - 1).$$
- ❑ Compute the public key e_2 using the formula:
$$e_2 = e_1^d \bmod p$$
- ❑ Thus, the **public key** is (e_1, e_2, p) , and the **private key** is d .

Encryption Process

- ❑ To encrypt a message M , the sender follows these steps:
- ❑ Convert the plaintext message M into a numerical form m (where $0 < m < p$).
- ❑ Choose a random integer r (a temporary secret key) such that $1 < r < p - 1$.
- ❑ Compute two ciphertext values:
- ❑ First part of the ciphertext: $c_1 = e_1^r \bmod p$
- ❑ Second part of the ciphertext: $c_2 = m \cdot e_2^r \bmod p$
- ❑ Send (c_1, c_2) as the encrypted message.

Decryption Process

- ❑ The receiver (who knows the private key) decrypts the ciphertext (c_1, c_2)
- ❑ Compute the shared secret using c_1 and the private key d : $s = c_1^d \bmod p$
- ❑ Compute the modular inverse of s , denoted as s^{-1}
(modular inverse exists because s is nonzero).
- ❑ Recover the plaintext message: $m = c_2 \cdot s^{-1} \bmod p$
- ❑ Convert m back to the original plaintext.



Example Calculation

Exercise: ElGamal Cryptosystem Implementation

Implement the **ElGamal public-key cryptosystem** step-by-step using the following parameters:

- **Prime modulus:** $p = 23$
- **Generator:** $e_1 = 5$
- **Private key:** $d = 6$
- **Random ephemeral key:** $r = 15$
- **Message to encrypt:** $m = 10$

Key Generation

Step 1 — Key Generation

Given:

- $p = 23$ (prime)
- $e_1 = 5$ (primitive root modulo p)
- Private key $d = 6$

So:

$$e_2 = 8$$

Public key: $(e_1, e_2 = 5^d \bmod p = 8)$

Private key: $d = 6$

We compute the public key e_2 :

$$e_2 = e_1^d \bmod p = 5^6 \bmod 23$$

First, calculate 5^6 :

$$5^2 = 25 \equiv 2 \bmod 23$$

$$5^4 = (5^2)^2 \equiv 2^2 = 4 \bmod 23$$

$$5^6 = 5^4 \times 5^2 \equiv 4 \times 2 = 8 \bmod 23$$

Encryption

Step 2 — Encryption

Given:

Plaintext $m = 10$

Random $r = 15$

$$5^8 \equiv 16$$

$$16 \times 4 = 64 \equiv 64 - 2 \times 23 = 64 - 46 = 18$$

Compute c_1 :

$$18 \times 2 = 36 \equiv 36 - 23 = 13$$

$$c_1 = e_1^r \bmod p = 5^{15} \bmod 23$$

$$13 \times 5 = 65 \equiv 65 - 2 \times 23 = 65 - 46 = 19$$

We can use modular exponentiation
step-by-step:

$$5^2 \equiv 2$$

$$5^4 \equiv 2^2 = 4$$

$$5^8 \equiv 4^2 = 16$$

So:

$$c_1 = 19$$

$$\text{Now } 5^{15} = 5^8 \times 5^4 \times 5^2 \times 5^1$$

$$\text{Check: } 15 = 8 + 4 + 2 + 1$$

Encryption

Compute c_2 :

$$c_2 = m \times (e_2^r) \bmod p$$

First find $e_2^r = 8^{15} \bmod 23$:

We know from the example

hint: $8^{15} \bmod 23 = 2$.

Let's verify briefly:

We can notice $8^2 = 64 \equiv 64 - 2 \times 23$
 $= 64 - 46 = 18$

Better approach:

$$8^2 \equiv 18$$

$$8^4 \equiv (18)^2 = 324$$

$324/23 \approx 14.086 \rightarrow 23 \times 14 = 322$,
remainder 2 $\rightarrow$ So $8^4 \equiv 2 \bmod 23$

So $8^{15} = 8^{12} \times 8^3 \equiv 8 \times 8^3$ — let's check exponent properly.

Actually $15 = 4 + 4 + 4 + 3 \rightarrow$ not neat. Let's do direct power chain:

Since $8^4 \equiv 2$,

$$8^5 = 8^4 \times 8 \equiv 2 \times 8 = 16$$

$$8^{10} = (8^5)^2 \equiv 16^2 = 256$$

$$256/23 = 11 \text{ remainder } 3 \rightarrow 8^{10} \equiv 3$$

$$8^{15} = 8^{10} \times 8^5 \equiv 3 \times 16 = 48 \equiv 48 - 46$$
$$= 2 \bmod 23 \quad \checkmark$$

So $8^{15} \bmod 23 = 2$.

Thus:

$$c_2 = m \times (8^{15}) \bmod 23 = 10 \times 2 \bmod 23 = 20$$

Ciphertext: $(c_1, c_2) = (19, 20)$

Decryption

Private key $d = 6$,

received $(c_1, c_2) = (19, 20)$

Compute shared secret s :

$$s = c_1^d \bmod p = 19^6 \bmod 23$$

First, $19 \bmod 23 = -4 \bmod 23$ maybe easier, but let's compute directly:

$$19^2 = 361, 361/23 = 15 \text{ remainder } 16 \rightarrow 19^2 \equiv 16$$

$$19^4 = (19^2)^2 \equiv 16^2 = 256 \equiv 256 - 11 \times 23 = 256 - 253 = 3$$

$$\text{Now } 19^6 = 19^4 \times 19^2 \equiv 3 \times 16 = 48 \equiv 48 - 2 \times 23 = 48 - 46 = 2$$

So: $s = 2$

Find modular inverse of $s = 2 \bmod 23$:

We need s^{-1} such that $2 \times s^{-1} \equiv 1 \bmod 23$.

Trial: $2 \times 12 = 24 \equiv 1 \bmod 23$ ✓

Thus $s^{-1} = 12$.

Decryption

Recover m :

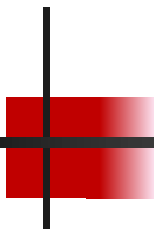
$$\begin{aligned}m &= c_2 \times s^{-1} \bmod p \\m &= 20 \times 12 \bmod 23 \\20 \times 12 &= 240\end{aligned}$$

$$240/23 = 10 \text{ remainder } 240 - 230 = 10$$

So:

$$m = 10$$

✓ **Original plaintext $m = 10$ recovered.**



Exercise

Exercise 1 – Complete Example

Given:

- $p = 19, e_1 = 2, d = 5, r = 7, m = 8$

Exercise 2 – Error Detection

Scenario:

- $p = 17, e_1 = 3, d = 4$

- Received ciphertext: $c_1 = 6, c_2 = 11$

- Bob computes $s = 13, s^{-1} = 4$